ADITYA C. DOPPALAPUDI

Dallas, TX | 757-754-6348 | aditya.doppalapudi@gmail.com | U.S. Citizen
https://www.linkedin.com/in/adityadoppalapudi/

MS Computer Science graduate with hands-on experience in Generative AI, LLM security, cloud-based AI workflows, Python development, and data analytics. Built secure code generation systems using Google Cloud Vertex AI and CodeQL, reducing vulnerability rates across 1,000+ generated code samples. U.S. Citizen with no sponsorship requirement. Seeking Software & Data Engineering and Full Stack Development opportunities.

EDUCATION

Old Dominion University — Norfolk, VA
Master of Science in Computer Science (MSCS) – GPA 3.97 — Graduated: May 2026
Relevant Coursework: Generative AI; Data Science; Advanced Databases; Big Data; Quantum Computing
Jawaharlal Nehru Technological-UCEST — Hyderabad
Bachelors of Technology in Electronics and Communication Engineering (ECE) — Graduated: July 2024

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Python, SQL, MySQL, Java, MATLAB, Bash, C++, HTML/CSS, RESTful APIs
Tools: Claude Code, Codex, GitHub, GIT, Linux, Google Cloud Vertex AI, GCP, CodeQL, ArcGIS, Docker
Certifications & Training: IBM: Basics of Quantum Information, Python for Everybody, Bash Shell Scripting, GIT Essentials, IBM: Intro to Practical Quantum-Safe Cryptography
Other: Generative AI, LLMs, RAG, Software Engineering, Static Analysis, Data Structures & Algorithms, SDLC

PROJECTS

LLM-Based Secure Coding Assistant | Python, CodeQL, Google Vertex AI, LLMs — December 2025

- Developed an **LLM-based secure code generation pipeline** using Python and CodeQL, implementing both **Proactive Vulnerability Prevention** and **Post-Hoc Vulnerability Repair** techniques.
- Designed and implemented a **CWE-based vulnerability hint generator** to guide LLMs toward safer code generation through security-focused prompting.
- Engineered an automated **static analysis feedback loop** using CodeQL findings and LLM-generated explanations to detect, explain, and repair vulnerable code which is integrated into the pipeline.
- Evaluated the pipeline across **1,071 code samples** using **TarV-R and AllV-R metrics**, achieving approximately **65% lower AllV-R vulnerability rate** with Google Cloud Vertex AI-based LLM workflows.

Movie Recommendation Engine | Python, PySpark, Spark MLlib, Pandas

- Developed a collaborative-filtering movie recommendation system using PySpark and ALS, processing 100,000+ MovieLens ratings and implementing data preprocessing, train-test splitting, and model evaluation.
- Implemented cross-validation and RMSE-based evaluation, achieving ~0.89 test RMSE and generating personalized top-5 movie recommendations with predicted ratings and movie metadata.

WORK EXPERIENCE

Old Dominion University (ODU) — Norfolk, VA
Graduate Teaching Assistant — January 2025 - May 2025

- Facilitated study sessions and grading of assignments/exams for Discrete Structures and Computer Architecture, along with helping students understand logic, and architecture concepts.

Institute of Electrical and Electronics Engineers - Circuits and Systems Society (IEEE-CAS)
VLSI Intern — October 2023 - December 2023

- Improved hardware implementation of RSA Algorithm in FPGA to result in smaller delays and improve efficiency.
- Improvement in Rivest-Shamir-Adleman (RSA) algorithm, using Kogge stone adder (KSA) and Montgomery multiplier.
- Implementation of 4-bit KSA reduced the delay by 3X and the use of Montgomery Multiplier improved efficiency by ~34%.

LEADERSHIP & STUDENT ENGAGEMENT

Association for Computing Machinery [ACM] - Student Chapter of ODU — Event Coordination Officer from 2024-2026
Global Student Friendship Group — Student Leader for Outreach from Oct 2024- May 2026
Institute of Electrical and Electronics Engineers — Member Since 2022

Open to Relocation